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FORMAL AND INFORMAL DETERRENTS TO DOMESTIC VIOLENCE: THE DADE COUNTY SPOUSE ASSAULT EXPERIMENT*

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Recent formulations of deterrence theory suggest that informal sanctions mediate the effects of formal sanctions. A randomized field experiment in Dade County was designed to test the relative effectiveness of different police responses to incidents of spouse abuse. We test whether the deterrent effect of arrest interacts with the informal sanctions implied by employment status and marital status. The results indicate that, overall, formal arrest has no effect on occurrence of a subsequent assault. This overall lack of effect masks two offsetting effects: Arrest has a statistically significant deterrent *effect among employed suspects, whereas arrest leads to a significant* increase *in subsequent assaults among unemployed suspects. The interaction effect between arrest and employment status was also statistically significant. There were no significant effects associated with marital status. The interaction effect between arrest and a composite measure of "commitment" based on both employment and marital status was also statistically significant.*

The Minneapolis Domestic Violence Experiment examined the relative effectiveness of various police responses to cases of misdemeanor spouse assault (Sherman and Berk 1984a, 1984b). Cases were assigned to three standard methods police use: arrest, attempting to counsel both parties, and sending the suspect away from home for several hours. Based on arrest records and interviews with the victims, the results indicated that cases assigned to the arrest response had the lowest level of recidivism, supporting the hypothesis that arrest has a specific deterrent effect.

Largely because of these results, several police departments adopted a preferred (or mandatory) policy of arrest for cases of spouse assault (Sherman and Hamilton 1984; Sherman, Cohn, and Hamilton 1985; Cohn and Sherman 1987). One legal scholar used the Minneapolis study to justify a proposed model statute requiring police officers to make arrests in cases of misdemeanor domestic assault (Lerman 1984). Furthermore, based on the Minneapolis study, the Attorney General's Task Force on Family Violence (1984, p. 104) recommended that law enforcement agencies establish arrest as the preferred response in cases of family violence.

The Minneapolis experiment has generated considerable debate and criticism (Lempert 1984; Binder and Meeker 1988, forthcoming; Elliott 1989; Meeker and Binder 1990). Critics argued that further studies were needed before making major policy changes based on the Minneapolis results. As a result, the National Institute of Justice funded "replications" of the Minneapolis Domestic Violence Experiment in six additional cities: Atlanta, Georgia; Charlotte, North Carolina; Colorado Springs, Colorado; Dade County (Miami), Florida; Milwaukee, Wisconsin; and Omaha, Nebraska. These studies addressed the criticisms leveled against the Minneapolis study.

The results from five of these replications are now available. Researchers in three sites — Omaha (Dunford, Huizinga, and Elliott 1990), Charlotte (Hirschel, Hutchison, Dean, Kelley, and Pesackis 1990), and Milwaukee (Sherman et al., "From Initial Deterrence," 1991) — found no support for an arrest effect with respect to either official records or interviews with victims conducted six months after the presenting incidents. The Colorado Springs study found no deterrent effect of arrest when the analysis was based on official records, but found a strong deterrent effect when victim interviews were analyzed (Berk, Campbell, Klap, and Western 1991, 1992).

The Metro-Dade experiment found several statistically significant deterrent effects of arrest

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based on both victim interviews and official records (Pate, Hamilton, and Annan 1991). Based on interviews with victims within weeks of the presenting incident, for example, arrest was associated with reductions in the occurrence and number of subsequent assaults by the same suspect on the same victim. Interviews with victims six months after the presenting incident revealed significant deterrent effects attributable to arrest with respect to occurrence and time to a subsequent assault on the original victim; the effect on number of subsequent assaults was only marginally short of significance. Based on reports of subsequent arrests, cases randomly assigned to the arrest response had significantly lower occurrence rates and times to a subsequent assault than those assigned to the no-arrest response. The effect of arrest on subsequent assaults as recorded on official reports completed by officers after responding to calls for police service was not significant.

The conflicting results of the various replications have provoked considerable interest (Sherman 1992). Many of the attempts to explain these differences have suggested that formal sanctions may work differently depending on the mediating effect of informal sanctions (Sherman, Smith, Schmidt, and Rogan 1991; Sherman and Smith 1992; Berk et al. 1992). Even before the replication results were available, Sherman (1984b, p. 270) hypothesized that "more socially bonded people are more deterrable" and therefore employed persons and married persons would be more likely to be deterred by arrest.

FORMAL AND INFORMAL DETERRENTS

The theory of deterrence rests on the proposition that human behavior is to some degree rational, and thus can be influenced by incentives, particularly the negative incentives inherent in formal sanctions. This leads to the expectation that increases in the severity of penalties and the certainty of their imposition will discourage illegal behavior by increasing its perceived or threatened costs.

Zimring and Hawkins (1971) contended that the deterrent effect of formal sanctions may be enhanced if these sanctions are associated with informal sanctions — "for the majority of people the most degrading aspect of punishment is the social message it conveys" (p. 39). More recently, they argued that "official actions can set off societal reactions that may provide potential offenders with more reason to avoid conviction than the officially imposed unpleasantness of punishment" (Zimring and Hawkins 1973, p. 174). Tittle and Logan (1973) proposed that research on deterrence should consider the possibility that "formal sanctions can be effective only if reinforced by informal sanctions" (p. 386).

Elaborating on this idea, Williams and Hawkins (1986) broadened the conception of the deterrence process and posited that formal sanctions are augmented by the imposition of informal sanctions like *stigma* (e.g., embarrassment); *attachment costs* (e.g., loss of valued relationships); and *commitment costs* (e.g., loss of job or economic opportunities). Commitment costs arise from an individual's "stake in conformity" (Toby 1957; Briar and Piliavin 1965). Williams and Hawkins observed that if informal sanctions are activated by the formal sanction, then informal sanctions should be included in any account of the deterrent effect of formal sanctions.

Few studies have tested this broadened conception of deterrence as it applies to domestic assault. Berk and Newton (1985), alluding to the difficulties, stated:

> It is apparent that deterrence could be operating through a variety of mechanisms. Perhaps offenders are responding to the fear of further criminal sanctions. Perhaps offenders are responding to the implications of arrest for relationships with friends, family, and neighbors. Perhaps offenders are concerned about what an arrest will do to their employment prospects. . . . Until these and other possible mechanisms are disentangled, theories of deterrence are little more than rhetoric. (p. 262)

Recently, Williams and Hawkins (1989) explored the influence of perceived formal and informal sanctions on arrests for wife assault. They analyzed data from a national survey of married or cohabiting men to determine their perceptions of the likelihood of direct costs (e.g., arrest and going to jail) and indirect costs of committing spouse assault. Based on their earlier work, the authors defined indirect costs as "stigmatic costs (familial and self-humiliation), attachment costs (loss of partner or disapproval from friends and relatives), and commitment costs (loss of job)" (Williams and Hawkins 1989, p. 167). The results indicated that the various indirect costs contributed to a "general sense of fear about arrest" (p. 178) for wife assault. Because these findings were based on men's perceptions of a "hypothetical" arrest for wife assault, it is unclear whether an actual arrest for wife assault would generate similar findings, i.e., that informal processes of social control would influence the deterrent effect of formal sanctions.

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The Metro-Dade Spouse Assault Experiment offers an opportunity to test whether arrest is a more effective deterrent to further spouse abuse than alternative police interventions and to examine whether the deterrent effect of arrest is mediated by informal sanctions. We use the data collected during this experiment to examine the effects of two measures of informal control — employment status and marital status — as they interact with the formal sanction of arrest for spouse assault. Three hypotheses on the deterrent effect of arrest for spouse assault are tested: (1) the effect is greater among suspects who are employed (i.e., risk job loss) than for those who are unemployed; (2) the effect is greater among suspects who are married (i.e., risk losing the partner or being humiliated in the eyes of the partner) than among those who are not married; and (3) the effect is greatest among suspects with the highest level of commitment (i.e., suspects who are both employed and married).

THE METRO-DADE SPOUSE ASSAULT EXPERIMENT

The Metro-Dade Spouse Assault Experiment was designed to test the deterrent effect of arrest in domestic violence cases. Conducted by the Police Foundation in cooperation with the Metro-Dade Police Department, the study ran from August 24, 1987 through July 16, 1989. The experiment utilized a randomization procedure to assign cases involving misdemeanor spouse battery that met the following criteria: (1) Probable cause for misdemeanor spouse battery existed, (2) no felony had occurred, (3) victim and subject were both on the scene upon the officers' arrival, (4) the victim was not in immediate danger, (5) the victim was a female 18 years of age or older, (6) the officer was not assaulted by subject or victim, and (7) there were no outstanding arrest warrants, injunctions, or criminal protective orders for victim or subject.

Because of Florida statutory requirements, for the first 13 months of the experiment eligible cases had to involve couples who were married or who had been married.

The Sample

The experimental sample included 907 cases. These cases were contributed by a 396 officers. Of the suspects, 36.2 percent were Anglo, 41.7 percent were black, and 22.2 percent were Hispanic. Fifteen percent of the suspects were 18 to 25 years old, 43.8 percent were 26 to 35 years old, 34.3 percent were 36 to 50 years old, and 7.0 percent were over 50 years old. Officers suspected alcohol or drug use by suspects in 69.5 percent of the cases. Approximately 29 percent of the suspects were unemployed at the time of the presenting incident. Most of the couples (78.7 percent) were married at the time of the presenting incident, 2.9 percent were separated, 2.0 percent were divorced, and 16.3 percent had a boyfriend/girlfriend relationship.

Randomization and Treatments

Eligible cases were randomly assigned by the department's Computer Assisted Dispatch system to an arrest or a no-arrest response. Among the 907 eligible cases, 815 (89.9 percent) of the cases received the treatment to which they were randomly assigned; of the 92 misassigned cases, 88 were randomly assigned to the no-arrest condition but actually received an arrest.[1]

Once probable cause was determined, the suspect was told that he was under arrest and read his Miranda rights. The arresting officer then searched the suspect for weapons and contraband and handcuffed the suspect.

Upon arrival at the county jail, the transporting officer and the prisoner(s) were registered and admitted to a holding area. Once admitted to the facility, the transporting officer gave the arrest form to the receiving officer, who searched the prisoner, entered his name in the booking log, obtained his fingerprints, and took photographs ("mug shots").

The prisoner was then taken by the transporting officer back to the booking area and put in the custody of the receiving officer. If the prisoner had not been transported from a district police station, the receiving officer determined whether to release the prisoner on a "promise to appear" (PTA) basis. If a PTA release was offered, the prisoner's personal property was returned and he was released. If a PTA release was not provided, or if the prisoner had been transported from a district station, the receiving officer gave the prisoner the opportunity to post bond. If bond was posted, the prisoner's property was returned and the prisoner released. Although there was wide variation in the time spent under arrest, the average suspect spent 14.6 hours under arrest.

[1] In an independent, second-stage experiment, randomly assigned suspects received attention from a specially trained unit of domestic violence detectives. That treatment is not discussed here.

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Table 1. Logistic Regression Coefficients for Regression of Occurrence of Subsequent Assault on Selected Independent Variables: Metro-Dade Spouse Assault Experiment, 1987 to 1989

Independent Variable	Model 1	Model 2	Model 3
Arrest	-.160	.706	.922
	(.228)	(.565)	(.541)
Prior violence	1.280***	1.263***	1.278***
	(.259)	(.263)	(.261)
Black	-.228	-.255	-.219
	(.255)	(.258)	(.256)
Hispanic	-.372	-.329	-.334
	(.326)	(.328)	(.327)
Married	-.118	-.343	—
	(.271)	(.363)	
Employed	-.232	.647	—
	(.243)	(.382)	
Commitment	—	—	.168
		(.245)	
Employed × arrest	—	-1.704***	—
		(.508)	
Married × arrest	—	.349	—
		(.544)	
Commitment × arrest	—	—	-.759*
		(.347)	
Constant	-1.965	-2.418	-2.466

* $p < .05$ *** $p < .001$

Note: Numbers in parentheses are standard errors; N = 907 for all Models.

Arrestees were notified by the clerk of the court of the time and place of arraignment, which occurred within 21 days of arrest. At this hearing, the judge could accept a plea of guilty and bind the case over for trial, or release the defendant if probable cause was lacking.

Data Collection

Data concerning subsequent assaults by the suspect against the same victim were gathered from the Domestic Violence Continuation Report form appended to all offense reports involving domestic violence of any type.[2] These data were coded to represent whether a subsequent assault was recorded, the number of such assaults, and the length of time that elapsed until the first subsequent assault.

Analysis

Although analyses of the number of subsequent assaults and elapsed time to a subsequent assault have been conducted, we present here the results of the analysis of whether or not a subsequent assault occurred within six months after the presenting incident. Because this is a binary dependent variable, the distribution is not continuous, and the error variance is not homogeneous (Myers 1990). As a result, ordinary regression analysis or analysis of variance are inappropriate (Amemiya 1981, 1985; Pindyck and Rubinfeld 1991, pp. 248–52). Various approaches to analyzing these types of data have been suggested (Greene 1990, pp. 661–95; Pindyck and Rubinfeld 1991, pp. 254–69). For ease of analysis and interpretation, our analysis uses logistic regression.

Three models were tested using the occurrence of subsequent aggravated battery as the outcome measure. Independent variables included arrest status, prior violence, ethnicity, marital status, and employment status. The prior violence variable was coded 1 if the suspect recorded prior assault involving the same victim and 0 otherwise. Ethnicity was represented by two dichotomous variables, labeled "black" and "Hispanic"; "Anglo" is the deleted category. The marital status variable was coded 1 if the victim and suspect were married at the time of the presenting incident and 0 otherwise. The employment status variable was coded 1 if the suspect was employed (full-time or part-time) at the time of the presenting incident and 0 otherwise. A "commitment" variable was coded 0 for suspects who were unemployed and unmarried, 1 for suspects who were either employed or married, and 2 for suspects who were employed and married.

RESULTS

The results of the logistic regression analyses are presented in Table 1. Model 1 reveals no statistically significant effect of arrest on the occurrence of a subsequent spouse assault. This reflects the similarity between arrested and not-arrested suspects in the occurrence of subsequent reported assaults (9.0 percent and 10.6, respectively).

Model 2 includes all the variables from Model 1 and adds interactions between arrest and employment status and arrest and marital status. The

[2] Interviews with victims were conducted soon after the presenting incident and six months after that incident. For comparability, only the data from the Domestic Violence Continuation Reports are analyzed here. However, analysis of the victim interviews produced similar results.

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Figure 1. Percent of Suspects With a Subsequent Assault by Employment Status and Arrest Status

results reveal a statistically significant interaction effect between arrest and employment status, but the interaction effect between arrest and marital status was not statistically significant. This suggests that the deterrent effect of arrest is influenced by the informal sanctions implicit in employment status. The nature of this interaction effect is revealed in analyses of the arrest effect within subcategories of suspects (data not shown). These analyses reveal that this overall lack of effect masks two significant offsetting effects: Arrest had a significant deterrent effect among employed suspects, whereas arrest led to a significant increase in subsequent assaults among unemployed suspects. Figure 1 shows these differential effects graphically. Among unemployed suspects, 7.1 percent of those not arrested had a subsequent assault compared to 16.7 percent of arrestees. Among the employed suspects, however, the results were reversed: 12.3 percent of those assigned to the no-arrest response had a subsequent assault compared to 6.2 percent of arrestees.

Model 3 substitutes the combined commitment variable for the employment status and marital status variables. Results reveal a statistically significant interaction effect between arrest and the composite measure of commitment, indicating that the strong mediating effect of employment status combines with the negligible effect of marital status to produce a moderate effect.

SUMMARY AND DISCUSSION

The results of this study provide general support for the hypothesis that the deterrent effect of the formal sanction of arrest for spouse abuse is mediated by the informal sanctions implicit in employment status. Among employed suspects, arrest had a statistically significant deterrent effect on the occurrence of a subsequent assault as recorded on the Domestic Violence Continuation Report forms. Among unemployed suspects, on the other hand, significant *increases* in subsequent assault were associated with arrest. There were no differences with respect to marital status.

Tests for interaction effects between arrest and employment status revealed significant differences, but the tests for interaction effects between arrest and martial status revealed no significant differences.

As suggested by Sherman (1992) and others, these findings raise serious concerns about the appropriateness and efficacy of universally mandated or preferred arrest for misdemeanor spouse abuse. If arrest deters only those who have something to lose (e.g., a job), that fact must be taken into account when policies are established.

The discrepant findings produced by the various replications of the Minneapolis Spouse Abuse Experiment emphasize the need for further research. When investigating the effectiveness of deterrence strategies, attention must be focused on both formal and informal sanctions and the interactions between them.

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